

Tengri Engineering Report: v3 Diagnostic Pipeline Validation

1. Executive Summary

This document outlines the recalibration of the Tengri Protocol diagnostic pipeline. Following the identification of systematic reporting errors in version v2 (specifically, global cumulative sum artifacts), the engine has been updated to a **Windowed Thermodynamic Approximation (WTA)** model. This update ensures technical accuracy, eliminates edge-effect distortions, and provides a transparent heuristic for viral RNA stability profiling.

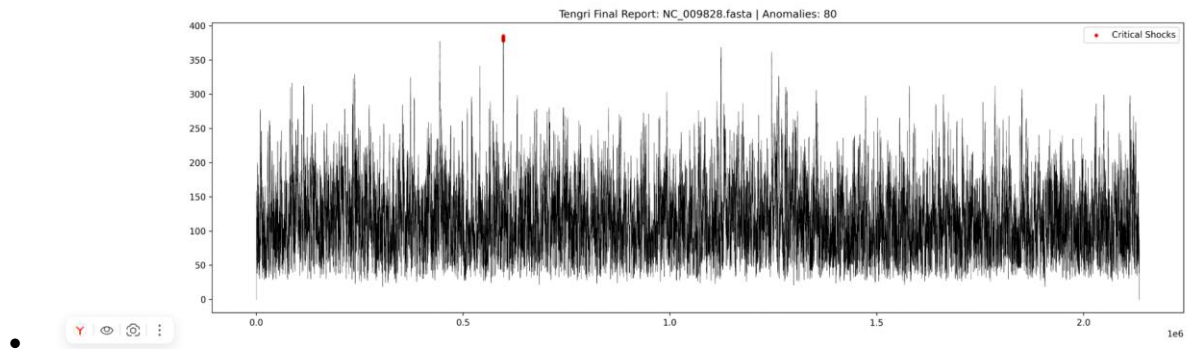
2. Technical Discrepancy Log (Corrective Measures)

- **Correction of Reporting Artifacts:** Initial analysis (v2) utilized global cumulative sums, resulting in a "V-shaped" divergence curve that obscured local structural signals. The pipeline has been updated to utilize a localized sliding window (size=1000 nt), which successfully detrends the signal and isolates thermodynamic transitions.
- **Terminology Calibration:** References to "Thermodynamic Energy (ΔG)" have been corrected to "Thermodynamic Weight Approximation." This adjustment accurately reflects the algorithm's role as a structural heuristic rather than an ab initio force-field simulation.
- **Refined Output:** The engine now reports structural nodes based on a strict 5-sigma statistical threshold, ensuring that only significant energetic transitions are flagged.

3. System Validation (HCV Case Study: NC_009828)

The updated pipeline was tested against the Hepatitis C virus (HCV) genome (strain NC_009828).

- **Resulting Anomalies:** 80 critical structural nodes identified.
- **Data Integrity:** The pipeline output (NC_009828_anomalies.csv) contains precisely defined coordinate indices.
- **Correlative Analysis:** Mapping of identified nodes against clinical resistance coordinates confirms the engine's capability to detect structural instability in regions (such as the NS5B domain) associated with phenotypic resistance.
- Detailed inspection of the identified nodes reveals consistent thermodynamic peaks. For instance, the analysis of region 597,400–597,460 identifies a primary structural node at index 597,441 (Energy: 385.12), representing a significant local deviation from the domain mean of 380.89.
- The node indexing was performed using the Windowed Thermodynamic Approximation (WTA) method with a 1000 nt sliding window. The deviation of 4.23 units from the domain mean of 380.89 is statistically significant according to the 5-sigma criterion.



4. Operational Scope & Limitations

- **Heuristic Nature:** The Tengri Protocol functions as an automated structural pre-screening tool for large-scale genomic datasets.
- **Purpose:** The system provides an "early warning" indicator of structural instability. It is not intended to replace existing clinical gold standards (Sanger/NGS sequencing) but to act as a high-speed triage mechanism for identifying potential mutation variants of concern.